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Naive algorithm for Pattern Searching

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Given a **text** string of length **n** and a **pattern** of length **m**, find all starting indices where the pattern occurs in the text.

Note: You may assume that $n > m$.

Examples:

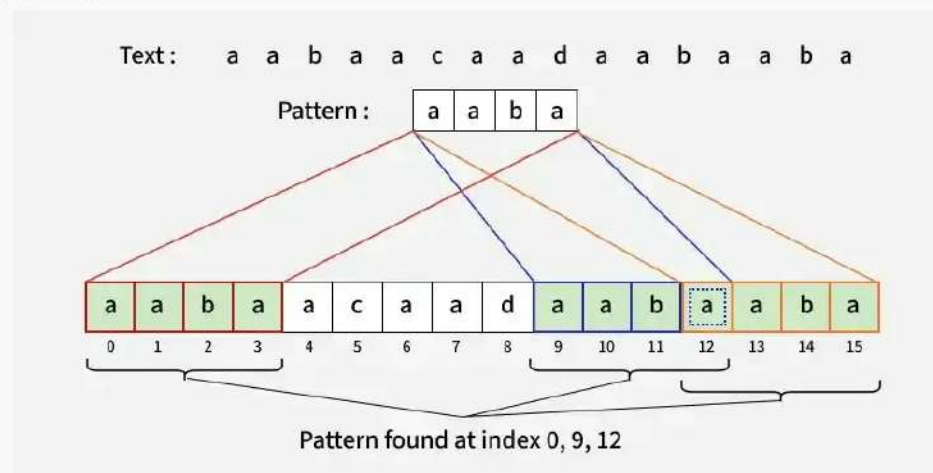
Input: text = "geeksforgeeks", pattern = "geeks"

Output: [0, 8]

Explanation: The string "geeks" occurs at index 0 and 8 in text.

Input: text = "aabaacaadaabaaba", pattern = "aaba"

Output: [0, 9, 12]



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```
import java.util.ArrayList;

public class GFG {

    // Function to search for all occurrences of 'pat' in 'txt' using Naive Approach
    static ArrayList<Integer> search(String pat, String txt)
    {
        // Length of the pattern
        int m = pat.length();

        // Length of the text
        int n = txt.length();

        ArrayList<Integer> ans = new ArrayList<>();

        // Slide the pattern over text one by one
        for (int i = 0; i <= n - m; i++)
        {
            int j;

            // Check for pattern match at index i
            for (j = 0; j < m; j++)
            {
                if (txt.charAt(i + j) != pat.charAt(j))
                    break;
            }

            // If pattern matches, store index
            if (j == m)
                ans.add(i);
        }

        return ans;
    }
}
```

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```
ArrayList<Integer> res = search(pat, txt);
```



```
for (int it : res)
{
    System.out.print(it + " ");
}
}
```

0 9 12

Output

0 9 12

Best Case Scenario

The best case occurs when the first character of the pattern does not match any character in the text.

Example: txt = "AABCCAADDEE" , pat = "FAA" . Only one comparison is done at each position. Time Complexity: $O(n)$

Worst Case Scenario

The worst case occurs when many characters match repeatedly.

Case 1: All characters are the same. Example: txt = "AAAAAAAAAAAAAAAAAAAA" , pat = "AAAAA"

Case 2: Only the last character is different. Example: txt = "AAAAAAAAAAAAAAAAAB" , pat = "AAAAAB"

In these cases, most characters match at every position, leading to repeated comparisons. Time Complexity: $O(m \times (n - m + 1))$

Time Complexity

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